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Governing rural electrification in Senegal: of monopolies, mini grids and beyond

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Abstract

Despite major progress in electricity access and renewable energy deployment over recent decades, Senegal continues to face challenges in achieving universal rural electrification, a country where 55% of the rural population lack access. A core issue is the absence of a coherent regulatory framework for rural electrification. Initiatives are fragmented, often pursue competing priorities, and involve a complexity of different actors and institutions. Within this context, mini grids have been promoted by donors and international organisations as a promising solution, and Senegal was once considered a regional leader in this sector. Today, most planned projects are solar or solar–diesel hybrid, as compared to the original diesel-generated systems. However, not only are a significant portion of the previously installed projects presumed to no longer be operational, but their actual and projected contribution to Senegal's rural electrification rate is relatively small. In the case of mini grids more specifically, there have been inconsistencies with regards to licensing, tariff-setting and arrangements for the arrival of the main grid. Processes and standards for their installation, operation, maintenance and ownership have been carried out in a somewhat haphazard way. Despite the introduction of regulation intended to support an increased role for the private sector in the electricity sector more generally, most mini grids that have been developed to date have been government-owned and donor-funded. Private sector involvement has been largely confined to engineering, procurement, and construction, and operation and maintenance. With this in mind, this paper critically examines the political, institutional, and regulatory barriers to rural electrification in Senegal. It highlights the tension between grid extension and the introduction of decentralised/off-grid systems, finding a significant mismatch between donor ideals and expectations on the one hand and the preferences of the state utility, as well as national and local governments on the other.

1. Introduction

Following several decades of regular and prolonged electricity blackouts and load-shedding, accompanied by a rise in electricity tariffs and food prices (Havard 2018), since 2012 Senegal has seen significant developments in its economy and electricity supply. Recategorised by the World Bank from low-income to lower middle-income in 2011, the country's electricity access rate increased from 50% to 70% of its population between 2006 and 2021 (Our World in Data 2023). While Senegal's power generation is still heavily dependent on imported and expensive heavy fuel oil and diesel, it now hosts some of the largest renewable independent power producers (IPPs) in the region and is in eighth place on the continent in terms of installed MW per capita for solar PV (AFSIA 2023:22). The country's original target of 20% of renewable energy by 2017, set under its previous national economic development strategy, the Plan Sénégal Emergent (PSE) was reached in 2019, and a new target subsequently set for 30% by 2030. A gas to power programme using recently discovered domestic reserves is under development.

Despite such developments, one of the PSE's core objectives, to provide an electricity connection to all citizens by 2023 and universal access by 2025, has not been met. The country's institutional and regulatory frameworks for rural electrification have had mixed results to date and of its population of 18 million people, there is considerable disparity between an *urban* electrification rate of 97%, largely served by the state-owned electricity company S  n  lec, and a *rural* one of 55% (IEA 2023). Senegal has some of the highest electricity tariffs in SSA (Marks 2021), due in part to the continued vulnerability of the sector to oil price volatility which depends heavily on public subsidies to make up the shortfall.

Within this context, mini grids have been put forward as a promising solution to rural electrification, particularly by donors and international institutions such as ESMAP, which has argued that mini grids could serve approximately 30% of Senegal's population (ESMAP 2022:79). Based on publicly available resources at the time of writing the country had a total of 677 of installed projects (Ibid p32). A further 1200 are planned, most of which will be 'third generation' (Ibid p xxxiv) to refer to solar or solar-diesel hybrid, as compared to many of the country's original mini grids which were diesel-generated (Signify Foundation 2019:4). However, not only are a significant portion of the installed projects presumed to no longer be operational, but their actual and projected contribution to Senegal's rural electrification rate is relatively small.

As this paper explores, Senegal lacks a coherent regulatory structure for rural electrification given that initiatives are not always well-aligned, often have competing priorities and involve a complexity of different actors and institutions (Ba 2018, Jaglin and Guillou 2020, Trompette *et al* 2022, Sarr 2024). In the case of mini grids more specifically, there have been inconsistencies with regards to licensing, tariff-setting and arrangements for the arrival of the main grid. Processes and standards for their installation, operation, maintenance and ownership have been carried out in a somewhat haphazard way with at times parallel structures (Jaglin 2019). Despite the introduction of regulation intended to support an increased role for the private sector in rural electrification, most of the mini grids that have been developed to date in Senegal have been government-owned and donor-funded, with private sector involvement largely confined to engineering, procurement, and construction and operation and maintenance contracts (Power Africa 2019).

In this paper I ask: *what have been the key governance challenges to the realisation of rural electrification in Senegal?* In so doing, I respond to calls for a deeper understanding of the conflict and co-existence that exists between the established centralised electricity grid, and the evolution of decentralised and off-grid systems (Jaglin and Guillou 2020, Rothfu   and Boamah 2020). I contribute to three main bodies of overlapping literature. First, to the growing literature on rural electrification in Senegal, including the patchy and complex nature of its regulatory framework e.g. Ulsrud *et al* (2018) and Caille and Badji (2018), Ba (2018), Diouf and Miezani (2021), Trompette *et al* (2022), Etienne (2022), Guillou and Girard (2022). Second, to studies on the development of decentralised systems in sub-Saharan Africa that have problematised the nature of regulation and private sector investment in rural electrification and mini grids in particular e.g. Sesan *et al* (2024b), Ogeya and Lambe (2025), Gaiya (2025). The paper's third contribution is to work on the political economy of ownership in the electricity sector in various countries in sub-Saharan Africa (e.g. Godinho and Eberhard 2019), including Senegal (Sarr 2024) and more specifically the role of development and climate finance and private investment in utility-scale renewable energy (Apfel 2022, van den Bold 2022, Haag 2023). While this literature has given critical empirical insights into how electricity as a socio-technical system interacts and co-evolves with multiple political, economic and social interests, it has given greater attention to the centralised system of generation, transmission and distribution and the role of IPPs as opposed to decentralised systems.

The structure of the paper is as follows: section 2 presents the methodology. Section 3 puts recent socio-technical and governance developments in rural electrification within a global context. Section 4 gives a regulatory overview of Senegal's electricity sector, including of key institutions and of the recent processes of power sector reform. Section 5 examines the various different programmes under which mini grids have been deployed and operated in the country so far and their varying levels of success. To that end, section 6 summarises the key challenges identified, and section 7 concludes, finding a significant mismatch between donor ideals and expectations, and the national realities and preferences of the state utility, and national and local government institutions.

2. Methodology

The paper's findings are based on 21 semi-structured qualitative interviews carried out by the author in January and February 2023 and January 2024 in Dakar, Thies, Mbour and surrounding areas. A smaller number of online interviews were carried out during March and April 2023. Interviews were held in French and/or English with members of civil society, the solar PV industry, academia and government

representatives responsible for electricity and energy policy, including the rural electrification agency and the state-owned utility, S  n  lec. Interviewees and their institutional affiliations have not been identified in the interests of commercial sensitivity. In addition to consulting the relevant academic literature, the paper draws from an extensive desk-based review of the grey literature concerned with rural electrification and electricity governance in Senegal, particularly since the start of its liberalisation processes in 1998. Documents consulted originated from government and public institutions directly involved in the energy sector, notably the rural electrification agency, the regulator, the utility S  n  lec, and the Ministry of Petrol and Energy (MPE); development finance institutions; international energy institutions; and national and international media; and quantitative data sources including IEA, IRENA and Our World in Data.

3. Governing rural electrification in context

The goal of Sustainable Development Goal 7 (SDG7) to ensure that everyone has access to affordable, reliable, sustainable, and modern energy by 2030 is well-rehearsed. In tandem, measurements and definitions of electricity access have shifted over the last decade, to which end, the World Bank's (2015) Multi-Tier Energy Access Framework (MTAF) has helped to provide a more nuanced understanding (see table 1). By categorising access into tiers based on the quality and quantity of electricity services available, the framework acknowledges that access is multidimensional and varies across different levels of service in both rural and urban settings. Despite this, conventional measurements still dominate global statistics and there is considerable nuance in what having an electricity connection might mean, for whom, at what cost and how it is measured (Ulsrud 2020). In some countries for instance, once the grid infrastructure reaches a village, national statistics assume that all those living there are connected. However, this assumption may not necessarily hold in the case of Senegal due to the prohibitive cost of a connection for many low-income households.

Rural electrification policies and regulation have evolved over recent decades alongside broader trends of power sector reform and rapid technological changes, not least the decentralisation of renewable electricity generation at multiple scales (Sareen *et al* 2024, Sesan *et al* 2024a). In the first half of the twentieth century the electricity sector, in countries where it was established, was generally a state-owned, vertically integrated monopoly. However, by the 1980s and 1990s, the 'standard model' of power sector reform, promoted by the World Bank, related consultants and other multi-lateral lending institutions as part of structural adjustment programmes was established as a global blueprint (Baker *et al* 2022). Informed by the experiences of a small group of mostly high-income countries (the USA, UK, Norway and Chile), the model proposed that state-owned electricity utilities should be unbundled into private generation, distribution and transmission companies, with a significant role for private investment, wholesale markets and ultimately, retail competition (Gratwick and Eberhard 2008; Sen 2014). The creation of a strong independent regulator to 'regulate the monopoly prone parts' of the industry was also prescribed (Victor and Heller 2007:7).

Despite the apparent simplicity of the standard model's logic, its implementation has been much more complex in practice, resulting in a diversity of failures and partial implementation (Patterson 1999:16). Reasons for this are numerous but include the inconvenience of political resistance by the state-owned utility and associated factions, and because the model simply did not fit the 'economic preconditions' (Ibid) of the electricity sectors of many low and middle-income countries. In addition, national regulators have rarely had the bargaining power anticipated by the model and have struggled to negotiate the complex contractual and regulatory terms of liberalisation with the government and other institutions. As the World Bank now acknowledges three decades on, only a dozen developing countries have been able to fully implement the standard model and it is generally accepted that a more flexible, and context-specific approach to power sector reform is needed (Foster and Rana 2020). Moreover, such reforms did little to address the reduction in energy poverty or increase rural electrification (Sokona *et al* 2012:3).

While the standard model was originally designed with the centralised electricity system in mind in reflection of the infrastructure of the day, the creation of rural electrification agencies and rural electrification funds were introduced from the 1990s as part of a 'second wave' of power sector reforms. As discussed below, Senegal, alongside Uganda, is an exception in that it also attempted to apply the model to rural electrification through the tendering of rural concessions to the private sector. In so doing, it encountered challenges in trying to establish a balance between financial viability and an affordable tariff, particularly for low-income users (Foster and Rana 2020).

While many principles of electricity liberalisation remain influential, decentralised and off-grid systems, including mini grids, have continued to evolve. These evolutions have been accompanied by

Table 1. Multi-tier framework electricity access tiers. Reproduced from Bhatia and Angelou (2015). CC BY 3.0.

	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5
Indicated minimum technology or system	Pico-products e.g. solar lamp and lantern; nano-grid; micro-grids; entry level SHS.	Basic capacity SHS to higher capacity SHS, rechargeable batteries, mini grids	High capacity SHSs, generators, mini grids, national grid	Mini grids, national grid	Mini grids, national grid
Power capacity ratings in Watt	Min 3 W	Min 50 W	Min 200 W	Min 800 W	Min 2 kW
Power capacity ratings in daily Watt hours	Min 12 Wh	200 Wh	Min 1.0 KWh	Min 3.4 kWh	Min 8.2 kWh
Availability hours per day	Min 4 h	Min 4 h	Min 8 h	Min 16 h	Min 23 h
Availability hours per evening	Min 1 h	Min 2 h	Min 3 h	Min 4 h	Min 4 h
Examples of appliances connected	2 lights, mobile phone charging	4 lights, mobile phone, radio	4 lights, phone, radio, TV, sewing machine, fridge, fan, computer	High-load appliances (washing machines, irons)	High-load appliances (vacuum cleaners, air conditioners, electric cookers)

Source: Bhatia and Angelou (2015).

shifting trends in global investment and technological innovation; changes in national electricity policy, planning and regulation; and the emergence of new private sector actors and the reconfiguration of state-owned utilities. Significantly, developments in decentralised and off-grid renewable systems mean that electricity can now be provisioned without the centralised mediation of the transmission or distribution grid. As a result, debates over the governance of electricity have gone beyond the simplified dichotomy of state versus market to include competition and co-existence ‘between different scales and configurations of centralised and decentralised electricity’ (Baker *et al* 2021:103). This has led to a new ideological dichotomy that pits the potential of a privately-owned and operated decentralised renewable sector against that of an indebted, capital-intensive, inefficient, vertically-integrated, state-owned utility, supplying an increasingly expensive and/or unreliable source of electricity (Trimble *et al* 2016). Such a discourse has been accompanied by an assumption that only the ‘dynamism of the private sector’³ can push forward the roll out of renewable electricity generation at scale, including from off-grid systems. However, as others have critically unpacked, efficient regulation alone is insufficient to encourage private sector involvement, to which the case of mini grids is no exception (cf Sesan *et al* 2024b).

The term ‘mini grid’ refers to a low-voltage or medium-voltage generation and electricity distribution network which can operate in isolation from the main grid but also connect to it should the network expand (Fajardo *et al* 2025). While mini grids differ in terms of technology, size, scale, ownership and customer base, they essentially allow power to be generated much closer to the point of consumption as compared to electricity via the centralised system of generation, transmission and distribution. ESMAP defines a mini grid as an electric power generation and distribution system that can ‘provide electricity to just a few customers in a remote settlement or bring power to hundreds of thousands of customers in a town or city. They can be fully isolated from the main grid or connected to it but able to intentionally isolate (“island”) themselves from the grid’ (ESMAP 2022:xxv). While some countries, including Nigeria, Kenya and Tanzania are now being celebrated for their mini grid regulatory frameworks (SEforAll *et al* 2024) the current and projected contribution of mini grids to rural electrification in Senegal is now relatively small and has been undermined by inconsistently applied policies and other challenges to their implementation and economic viability.

Finally, while mini grids have faced numerous policy and regulatory challenges, solar home systems (SHSs) by comparison have been implemented with greater ease and dynamism, partly because

³ Romain Py, Chief Investment Officer, Africa Infrastructure Investment Managers, South Africa, Africa Utilities Week, 11 May 2020 (publicly available conference).

they have often fallen under the radar of national regulatory requirements or electricity planning due to their small-scale standalone nature. Despite the potential and actual contribution offered by SHSs, they do so at a much more basic level of provision, often termed ‘pre-electrification’. Significantly, distributors of SHSs are not providing an electricity supply, but rather the technological kit. In the case of Senegal at least, SHSs have therefore faced fewer barriers to entry than those faced by mini grids (RVO 2022:23).

Against this global backdrop of shifting paradigms in electricity governance and technological decentralisation, Senegal’s approach to rural electrification, particularly through mini grids, offers a revealing insight into the mismatch between donor-driven templates, and national realities, regulatory fragmentation, and institutional inertia. These are illustrated most obviously in growing tensions between the state-owned utility, regional distribution companies, and new decentralised institutions and actors in the renewable energy space as I now describe.

4. Electricity in Senegal: a sector in transition

Senegal’s electricity sector is in transition, at once in terms of its ownership and governance, as well as its generation mix, which has gone from a capacity of 647 MW in 2014, made up almost exclusively of state-owned fossil fuel plants, to more than 1.5 GW in 2021 (RVO 2022:11, see figure 1). While just over 70% of installed generation capacity is from thermal sources, mainly heavy fuel oil and diesel, most of which is imported from Nigeria (ARDA, n.d.), renewable energy IPPs, mainly wind and solar PV, accounted for almost a quarter of installed capacity by 2022 and contributed an increasing share of actual generation—approaching 20% according to some estimates (see figure 1). Beyond this, approximately 66 MW and 15 MW comes from the Manantali and Felou hydroelectric power stations respectively, both located in Mali (République du Sénégal et MPE 2020:8). Other recent developments have included the updating and extension of various transmission and distribution lines, particularly between 2012 and 2016 (Gouvernement du Sénégal, n.d.). Beyond the centralised grid, an estimated 85 MW of capacity was from off-grid sources based on 2021 figures, largely SHSs and mini grids of which the proportion is unknown (Sénélec 2022:16, RVO 2022). There is also an emerging niche of small-scale private sector activities in the off-grid sector, including solar water pumps for irrigation and off-grid solar hubs for refrigeration and grinding mills but which do not make it into the electrification statistics (Potron 2024).

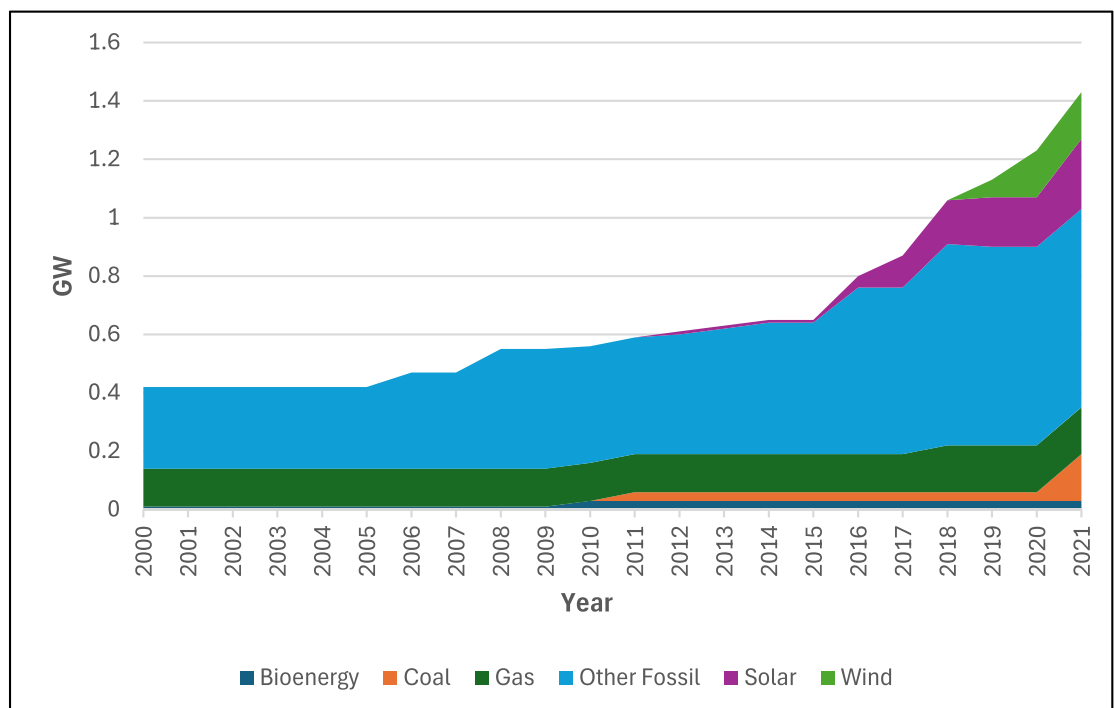


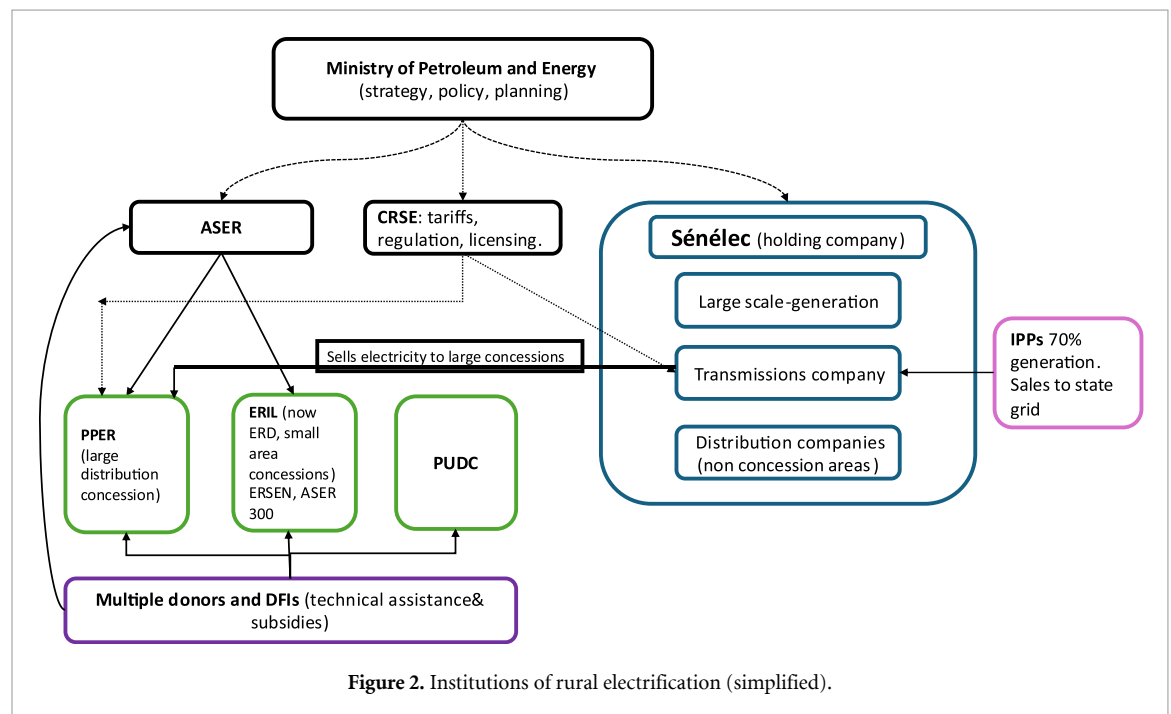
Figure 1. Senegal’s installed generation capacity 2000–2021. Reproduced from Ember (2024). CC BY 3.0.

The recent discovery of major oil and gas reserves, onshore in the south and offshore in the north of the country, has significant implications at once for electricity generation, its transmission and distribution grid infrastructure, and the economy more generally. Though much of the gas will be liquefied and exported to Europe, some of these reserves are slated to replace much of the heavy fuel oil and coal upon which Senegal's electricity sector still depends (République du Senegal 2021, Howard 2023) and are likely to have a significant influence over the future of the country's electricity and energy sector more generally (Sarr *et al* 2023).

Senegal's electricity sector is overseen by the Ministry of Petrol and Energy (Ministère du Pétrole et des Energies (MPE))⁴ which holds overall responsibility for strategy, policy and planning. The state utility Sénélec which had been nationalised in 1983, was then partially liberalised in 1998 following the standard model of power sector reform. The passing of the Electricity Law number 98–29 made Sénélec a limited company with a majority public shareholding (CRSE 1998). Though retaining the monopoly on the purchase, sale and transmission of electricity, the new law allowed for the procurement of electricity from IPPs with whom Sénélec is forced to compete, and which now account for 70% of generation (see figure 2). The new law also restricted Sénélec's right to distribute electricity to areas that were already electrified. As further explored below, areas of electricity distribution were opened to competitive tender in 2002 through concessionary contracts, with the aim of increasing the rural electrification rate to 60% by 2022 (Gesto Energia 2018:32).

Two key institutions were created from the 1998 law. First, the electricity regulator (La Commission de Régulation du Secteur de l'Électricité CRSE n.d.(b)), which is responsible for the overall regulation of the transmission, generation and distribution sectors; the sale of electrical energy, tariff-setting and licensing standards; the approval of revenue requirements; and leading the IPP tender processes (CRSE n.d.(b), République de Senegal 2010). Second, the rural electrification agency (L'Agence Sénégalaise d'Électrification Rurale, ASER), whose remit includes providing financial and technical assistance, managing rural electrification tenders and proposals from private sector renewable energy developers, overseeing installed mini grid systems, and the management of the rural electrification action plan discussed in section 5. The regulator, the rural electrification agency and Sénélec all fall under the supervision of the Ministry of Petroleum and Energy (see figure 2).

The 1998 Electricity Law was followed by moves to diversify the electricity mix. Notably, the adoption of the 2010 Renewable Energy Law, subsequent regulatory frameworks for the procurement, sale and purchase of renewable electricity generation, and a tax exemption for the purchase of renewable



⁴ Formerly, Ministère de l'Énergie et du Développement des Énergies Renouvelables (Ministry of Energy and the Development of Renewable Energy (MEDER)).

energy technologies for domestic electricity generation (République de Sénégal 2010). These objectives were operationalised in various policy documents, including the governing document of the energy sector, La Lettre de Politique de Développement du Secteur de l'Energie (LPDSE), originally released in 1997 but updated approximately every four years to reflect the content of the electricity law (République du Sénégal 2019). The 2012 version of the LPDSE was written to develop many of the objectives of the 2014–2018 version of the national economic development plan, the PSE, of which the energy sector was a priority and central to national efforts to secure emerging economy status by 2035. These objectives included the realisation of universal electricity access by 2025, the reduction of electricity tariffs, an overhaul of the generation mix, and the promotion of renewable electricity generation from IPPs (République du Sénégal 2014). The PSE's target for renewables to provide 20% of all power generation by 2017, and 30% by 2025 was achieved by the end of 2018. In 2013 a third institution was established, the national agency for renewable energy (L'Agence Nationale d'Energies Renouvelables, ANER), with a mandate to promote and create appropriate regulatory frameworks for renewable energy (ECREE 2018) but its impact has been limited.

While frameworks for rural electrification in Senegal have been patchy (Trompette *et al* 2022), regulation for the procurement and deployment of utility-scale renewable electricity generation from the private sector has been far more successful in terms of project completion (van den Bold 2022, Apfel 2022, Haag 2023). The country now has some of the largest utility-scale renewable IPPs in West Africa, including the 158 MW Taiba Ndiaye wind farm, and the Senergy and Ten Merina solar PV plants, with a generation capacity of 30 MW each. These two projects, owned by French utility Engie, private equity firm Meridiam and the local sovereign wealth fund, Fonds Souverain d'Investissements Stratégiques (Fonsis) were developed under the IFC's Scaling Solar programme and became operational in 2017 (BII n.d.).

The most recent major changes in Senegal's power sector liberalisation have come with the adoption of law 2021–31 on the Code de l'Électricité in July 2021, which restructures Sénélec into a holding company and creates three independent companies in charge of generation, transmission and distribution respectively thereby putting an end to its remaining monopoly over the purchase, sale and transmission of electricity. The code also aims to facilitate further private sector investment and participation in the sector; improve electricity planning and modelling; extend greater powers to the regulator; use national domestic gas finds to reduce electricity tariffs by 40%; and provide least cost electricity, particularly for rural populations by introducing a new model of rural electrification, as discussed in section 5.2 (République du Sénégal 2021). Further reform to the code was introduced in February 2023, setting conditions on the self-generation of electricity for domestic, industrial, commercial or industrial use below 1 MW in the case of solar PV and 2 MW for other generation sources. This, in addition to the sale of surplus electricity and its integration into the main grid (République du Sénégal 2023, CRSE 2023a). Allowing for electricity consumers to sell their own generation back into the grid, this reform is therefore considered a potential game changer.

4.1. Electricity access in context

As discussed above, there are significant regional disparities in terms of electricity access in Senegal (see figure 3). Twenty per cent of the population is concentrated in Dakar, which accounts for sixty per cent of national electricity demand. The population has an urban-rural distribution of approximately 50/50%, now gradually shifting towards the urban (Our World in Data 2024), while 37.8% of the population live below the poverty line, of which the majority in rural areas⁵ (World Bank 2023). The development of Sénélec's grid was originally focussed on supplying industrial centres, public lighting, and domestic users in the capital Dakar, and the four main urban areas of St. Louis, Kaolack, Ziguinchor and Tambacounda. The country is divided into 14 regions, with those in the west being much more densely populated than those in the east. The eastern and southern regions have some of the lowest rates of electrification, except for the city of Ziguinchor at 90% (see figure 4). Beyond Sénélec, half of the country's population lives within the electricity distribution concessions, discussed in section 5.1, where many of the rural poor either live a long way from the distribution grid or live 'under it', depending on traditional energy sources for lighting and cooking such as biomass, firewood, charcoal, candles and kerosene (Ba 2018:25) and/or low levels of electricity consumption, mainly for lighting and mobile telephone charging via SHSs.

The government's 2020 operational plan for the realisation of universal electricity access for all by 2025 and its accompanying investment prospectus proposed that of the nearly 14 000 rural villages still

⁵ Using the national poverty line (lower-middle income) of \$3.70 a day (2017 purchasing power parity) (% population): 38.40% in 2018.

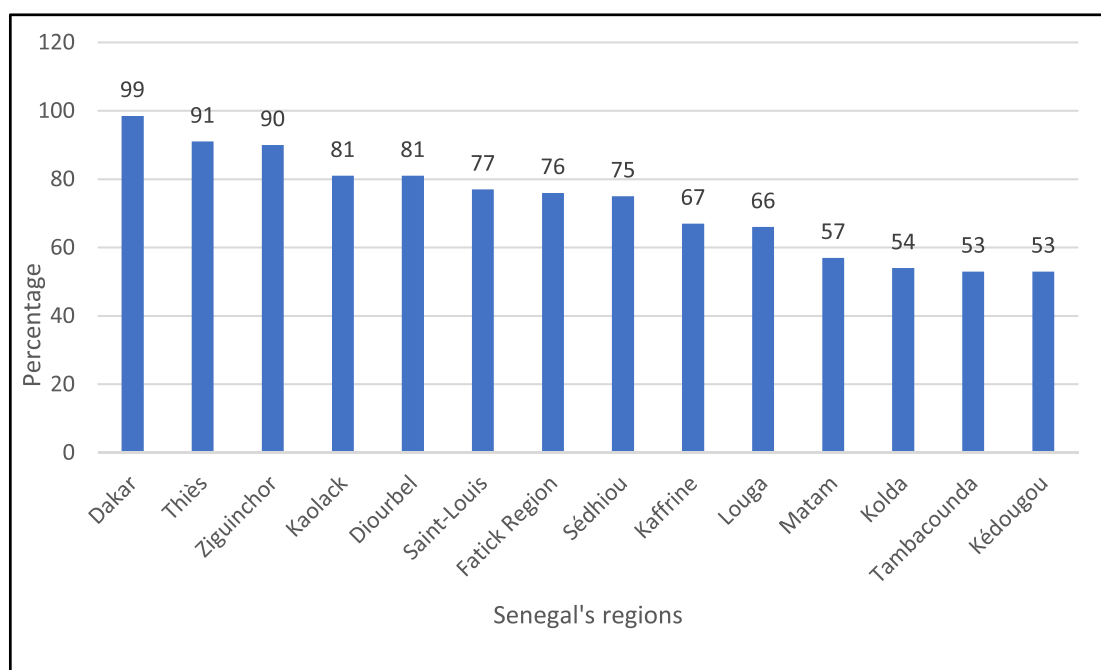
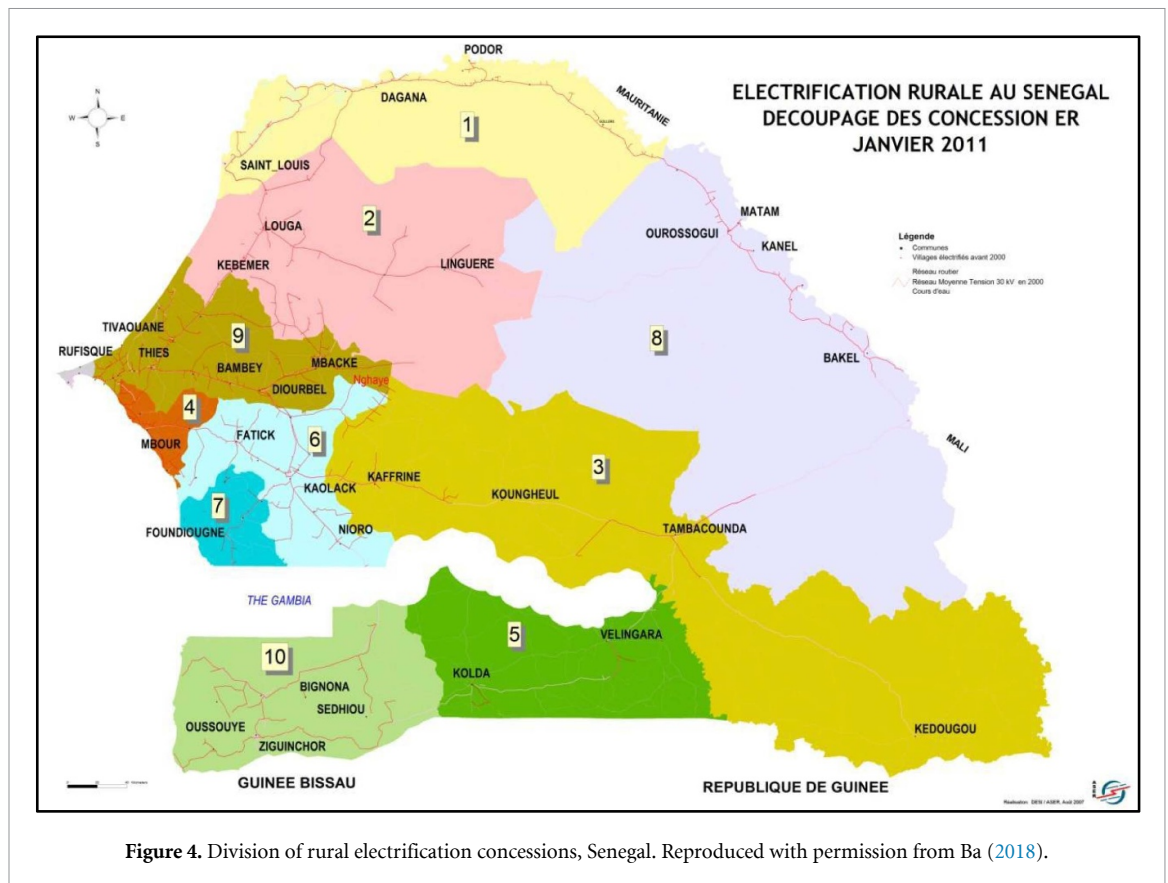


Figure 3. Senegal electricity access rates by region in 2022. Reproduced from (Agence Nationale de la Statistique et la Démographie (ANSD) 2024). CC BY 4.0.

to gain electricity access, a total of nearly three million habitants, over 50% will be electrified by extending the network, 32% by SHSs and only seven per cent by mini grids (MPE, ASER et Semis 2020a:8; République du Sénégal et MPE 2020). In household terms, this would mean 84% to be electrified by grid connection and eight per cent each by mini grids and SHSs (RVO 2022:23).

Senegal's rural population is dominated by small-scale agricultural producers who rely on what they can grow for their livelihoods, in addition to irregular seasonal income, such as fishing and/or livestock. They are therefore very vulnerable to external events such as drought or fuel price hikes. Communities are not necessarily static; individuals may go and work in urban areas for months at a time or leave the area during fishing seasons (Interview 8). During periods of low or no monetary income generation, households may depend on deferred payments for goods and services, payment in kind e.g. local produce, and loans and transfers from family or community members, particularly those working in urban areas or abroad (Interview 4). People may also simply 'accept "staying in the dark" when they are cashless' (Cholez and Trompette 2019: 348). Irregular monetary income in rural areas means that a regular economic subscription to an energy service, be that an SHS, a connection to a mini grid or even the main grid, is therefore a 'recurring expense that many households cannot cover' (Ibid p348). Low-income communities may also have limited if any experience of formal borrowing or debt repayment. During periods of high-income e.g. the harvest, households may wish to clear all their debts at once, as well as paying for services in advance thereby failing to conform to the monthly billing cycle. Because of the nature of its livelihood, this population has been defined as unreliable and unstable by the financial institutions associated with rural electrification, and neither government-run nor private sector operators have great incentive to electrify such populations without significant subsidy (DAPSA 2021, Trompette *et al* 2022).

In 2018 a process to harmonise the national electricity tariff was introduced, prior to which the cost of grid-connected electricity in rural areas in Senegal was at least two to three times more expensive than in urban areas. This meant that many electricity consumers on the lowest incomes were paying more than their metropolitan counterparts (Ba 2018, Cholez and Trompette *et al* 2019). As discussed below, while the harmonisation process has attempted to standardise electricity billing across the entire country, it has not been implemented consistently, particularly in areas of low electricity consumption. The outcomes are also double-edged: while it has the potential to make rural electrification more affordable, it may also decrease the profitability, and therefore the interest, of mini grids for investors and operators.



5. Rural electrification in Senegal

Following the global trends and subsequent national policy changes discussed in sections 3 and 4 respectively, two major concession programmes were introduced in 2002 as part of Senegal's rural electrification action plan. The aim of these concession programmes was to increase rural electrification from eight per cent in 2000 to 62 per cent by 2022 by encouraging private sector participation in off-grid and on-grid systems. First, through the tender process for rural distribution concessions under the priority rural electrification programmes (PPERs) (Gouvernement du Sénégal nd) and second, under local rural electrification initiatives based on small-scale concessions for local community associations (ERIL, Electrification Rurale d'Initiative Locale) (CRSE n.d.(a)) (see figure 2). Both programmes are technology neutral, meaning that there is no obligation for any particular technology to be used; both are managed by ASER; and both have been heavily subsidised by government and/or multi-lateral or bi-lateral donors. Though neither have performed well in terms of meeting rural electrification targets, the ERIL programme has been marginally more successful in terms of mini grid deployment than the PPER. Significantly, the anticipated private sector investment that such programmes were meant to encourage as part of the standard model of power sector reform discussed in section 3 has not materialised. In parallel to the ERIL and the PPER programmes there is an additional initiative for rural electrification called the PUDC with which there has been poor coordination. Each are now discussed in turn.

5.1. Top down: PPERs

Senegal was one of the first countries to pursue rural electrification on a significant scale through an area-based concession approach (World Bank 2004), following the removal of Sénélec's right to distribute electricity to non-electrified areas under the 1998 electricity law and the transfer of responsibility to private sector operators under the coordination of ASER. By 2012 the government had divided the country's rural regions into ten large concession areas across the country (see figure 4) and launched the PPER, which saw private consortiums of international developers and Senegalese companies bid to install and operate a mix of grid extension and off-grid electrification in the concession areas, taking responsibility for the generation, transmission, and distribution of the entire area (SEforAll and BNEF 2020:78, CRSE nd-a, see table 2). Successful bidders were selected based on the greatest number of households they proposed to supply, in return for a 25 year contract with CRSE, a pre-defined subsidy

Table 2. Rural concession companies and targets met under the PPER (based on 2019 data).

	Concessionary area	Company	Funder	Investment	Percentage subsidy	Entry into force	Infrastructure network percentage of target met ^a	Solar installations percentage of target met ^b	Household connection targets	Percentage of connection target met
1.	Dagana-Podor-Saint-Louis	Compagnie Maroco Sénégalaise d'Electricité (COMASEL)	World Bank	18.5 m	35%	2011	95%	68%	19 574	54%
2.	Louga-Kebemer-Linguere	Compagnie Maroco Sénégalaise d'Electricité (COMASEL)	AfDB	18.4 m	78%	2012	85%	108%	11 826	68%
3	Kaffrine-Tamba-Kedoudou	Energie Rurale Africaine (ERA). A JV between EDF (France) and Matforce (Senegal).	AfDB	15.8 m	64%	2014	35%	136%	18 001	52%
4	Mbour	Groupement Sénégalais Coselec et les Câbleries (SCL) Energie Solutions: Steg International Services Mbour SCL	World Bank	14.9 m	36%	2015	18%	0%	7,500	68%
5	Kolda-Vélingara	Groupement ENCO—ISO FOTON Maroc ^c (Isoton Spanish) (Enco-Senegalese subsidiary of a Senegalese company).	Senegal/EU	22.8 m	30%	2015	0%	0%	20 500	0%

(Continued.)

Table 2. (Continued.)

	Concessionary area	Company	Funder	Investment	Percentage subsidy	Entry into force	Infrastructure network percentage of target met ^a	Solar installations percentage of target met ^b	Household connection targets	Percentage of connection target met
6	Kaolack-Noro-Fatick Gossas	Groupe ment ENCO—ISO FOTON Maroc ^d (Isoton Spanish) (Enco-Senegalese subsidiary of a Senegalese company)	EU	15.9 m	69%	2014	0%	0%	27 000	0%
7	Foundiougne	Sénélec	N/A			2018				
8	Matam-Bakel-KanelRanerou	Sénélec	N/A			2018				
9	Rufisque-Thies-Tivaouane Bambe y-Diourbel Mbacke	Sénélec	N/A			2018				
10.	Ziguinchor - OussouyeBignona-Sédhiou;	Sénélec	N/A			2018				

Source: Author's own, compiled from SE4All & AfDB (2019:24); (World Bank, 2015):17; (USAID and Power Africa 2018, 2018):66; (Power Africa, 2019):3; (RVO 2022, 2022):22.

^a Medium and low voltage network, includes grid extension and mini grids.

^b Standalone systems or solar PV mini grids.

based on their generation output, and further subsidy for the installation of renewable energy technologies (Mawhood and Gross 2014). The PPER does not include small village-level operations, which have been tendered under the ERIL programme (section 5.2).

In terms of outcomes, bidders largely focussed on grid extension, to a lesser extent SHSs, and to an even lesser extent, mini grids (World Bank 2015). The PPER has had limited results in terms of meeting electricity access targets (see table 2) and faced numerous interrelated challenges, not least an overly complex institutional framework and significant delays in establishing the tender process and awarding the concessions. Notably, some concession areas were considered too large to be manageable, resulting in limited interest from the private sector. As a result, only six out of the ten areas were allocated to private companies (SEforAll and BNEF 2020:78, CRSE nd-a, see table 2), for which contracts eventually became operational between 2011 and 2015 (World Bank 2015). The remaining four were granted to S  n  lec in 2018, on whose behalf they are now managed by transitional management companies.

Concession areas have an average radius of 100 km (see figure 4), but differ widely in terms of geo-physical characteristics, socio-economic development, infrastructure coverage and access to electricity. For instance, the largest concession, Kaffrine-Tambacounda-K  dougou, covers approximately 25% of Senegal's landmass but includes some of the country's least densely populated areas (ERA Senegal n.d.). Regardless of income levels, the regions with the lowest population densities have the lowest electrification rates, while the electricity concessions with higher population densities have attracted higher levels of private investment (Diouf and Miezan 2021:630). With an overall collective target of 106 601 households for the six concession companies, individual targets range from 11 800 to 19 000 households per area based on available information (RVO 2022:22, see table 2). Only two of the six concessionaires had made any significant progress by 2018 (Ba 2018:33) and by December 2020 they had collectively connected only 46% of the overall target (IDA 2022:1).

The competitive bidding programme and the various tenders received significant financial and technical support from DFIs and bi-lateral donors, some of which was packaged jointly for both the PPER and the ERIL programmes (see e.g. World Bank 2013). The concessionaires subsequently received significant development finance (Power Africa 2019:3, see e.g. IFC 2011).

Despite the positive discourse surrounding the potential of mini grids for energy access and rural electrification, their contribution within the concession areas has been small and the process for their approval, installation and maintenance has not been straightforward. The process is as follows: the rural electrification agency (ASER) decides on the capacity and the location of the projects and launches a tender for private companies to install them. After their installation, ASER retains system ownership but launches another tender for the operation and maintenance contract (Power Africa 2019). The concessionaires have the right of first refusal to bid, but if they are not interested, or their bids are unsuccessful, a subsequent tender is launched. ASER then transfers the system operation to the successful bidder, with whom it signs a 25 year contract requiring the operator to provide services for at least six hours per day (World Bank 2015:23). According to interviewees, some operators have had to pay additional money for project maintenance due to the poor technical standards of installation. Meanwhile, S  n  lec's strategy within its four concessionary areas has largely focussed on grid extension and tended to overlook the energy needs of the highly dispersed rural population who present limited economic incentive (Diouf and Miezan 2021).

Prior to the tariff harmonisation process introduced in December 2018, concessionaires and any mini grid operators within the concessions were at liberty to set their own tariffs. These tariffs were nearly always higher than in S  n  lec's urban distribution areas, often by up to 25% (SEforAll and AfDB 2019:23), not least because concessionaires had often been charging consumers for the cost of connection and the installation of internal wiring in addition to their electricity consumption (Interview 8). Tariffs also varied between and within the different concessions, reflecting the very different conditions within each area. This situation changed with the harmonisation process which required that concessionaires reduce their tariffs, and that any mini grid project developed after 2018 would only be approved if it did not exceed the harmonised tariff.

Tariff harmonisation also led to a significant restructuring of the contracts between the concessionaires and ASER, as a result of which concessionaires are now eligible to apply for compensation from the Ministry of Finance via the CRSE on a tri-monthly basis to justify the reimbursement of their revenue shortfall (interviewee 3⁶). According to IDA, by 2022 this had incurred government compensation of \$50 million (IDA 2022) but such a process has been protracted, complex and much harder for smaller companies who do not have the capital to wait for reimbursement (Trompette *et al* 2022, Faye 2023).

⁶ See also for example CRSE (2023b), with respect to compensation paid to SCL Energies Solutions.

Though the aim of tariff harmonisation was to increase the share of the population able to pay for electricity (RVO 2022:22), its impact on connection rates has been difficult to assess. One reason for this is because since its introduction, concessionaires are no longer obliged to pre-finance, and therefore to record, the internal wiring of customers' houses as was the case before harmonisation (IDA 2022). According to various interviewees, tariff harmonisation has a positive potential for rural electrification from mini grids in that previously, there were many communities who would refuse to connect to a mini grid, stating a preference to wait for the distribution grid from S  n  lec or the concessionaire. As interview 3 explained 'people perceive [the main grid] as a better quality and more constant supply of electricity'. On the other hand, some mini grid operators have seen tariff harmonisation as a deterrent to profitability because many concessionaires lost the limited interest that they may have previously had, preferring instead to extend the distribution grid to a greater or lesser extent (interview 12).

5.2. Bottom up: Local rural electrification initiatives

Given the slow progress of the PPER towards meeting the 2025 access target (see table 2), the ERIL programme was introduced subsequently to focus on households in communities not covered by either the electricity distribution concessions or S  n  lec's network, and where there were no plans for grid electrification for the following three years (MPE and GIZ 2020). Under the ERIL programme, small and medium-enterprises, NGOs and private or public sector actors bid to supply electricity to local communities through the operation and maintenance of stand-alone third generation mini grids and SHSs, for which they receive subsidies from government and/or donor organisations. However, the programme's implementation has been problematic and its contribution to increased levels of rural electrification understood to have been marginal so far.

ERIL projects have either been set up by developers whose proposals were approved by ASER, as for instance in the case of the Dutch company Daey Ouwens which since 2011 has set up some of the few functional projects (interview 7, see table 3, RVO 2024), or operators who have more recently responded to ASER-run tenders (Takoueu 2021). Under these tenders, project operators are fully responsible for operation, maintenance and billing and required to adhere to the technical standards set by that tender (MPE and GIZ 2020:6). Nearly all operators are Senegalese SMEs, for example EnergieR, Faye Solaire, Sud Energie and SALENSOL (CRSE, nd-a; Power Africa 2019:19). The companies that collectively operate the largest portfolios of installed systems are NS RESIF (NSRESIF 2018) and S  d Solar System (Sud Solar System 2017, ESMAP 2022:157 and 32), having obtained approval from ASER to develop in multiple villages and operate various projects under the ERS  N programme (discussed in section 5.2.1).

As with the PPER, many of the ERIL tenders have been supported by various DFIs, with GIZ playing a particularly significant role through technical assistance, training and technological innovation (interview 3). ERIL projects are also heavily subsidised and can receive up to 80% of subsidy from the Senegalese state or DFIs (Trompette *et al* 2022:73, RECP *et al* 2014). Despite being designed to attract private sector investment, most ERIL projects have so far been entirely funded by bi-lateral or multi-lateral development institutions including the Islamic Development Bank, USAID and the West African Development Bank (RVO 2022, Etienne 2022, Islamic Development Bank 2020, see table 3). While the relative lack of data on the ERIL programme makes it hard to evaluate (RVO 2022:22), that which is available is presented below.

The various challenges attributed to the implementation of the ERILs have included: a weak mobilisation of private sector financing; the absence of a reliable data collection and accounting system between the operator and ASER, meaning that project use has not been well recorded or understood; inadequate systems for management and monitoring, which has affected the sustainable functioning of many projects; the malfunctioning of invertors and batteries due to extreme heat and dust; a lack of technical oversight, further undermined by the lack of trained personnel; poor security measures in the vicinity of each project which has resulted in equipment theft; and the lack of education for users regarding the capacity and capability of the system (MPE and GIZ 2020).

Moreover, a regulatory framework between local authorities and the institutional players involved in ERIL has been lacking (MPE and GIZ 2020:12). Most projects installed have been operating without licences or tariff reviews since their inception, with only one contract having been signed with CRSE in 2019, by the German-Senegalese joint venture, ENERSA which operates in Tambacounda (SEforAll and AfDB 2019:11). For this reason, the tariff harmonisation process discussed above has not yet been applied and for the time being, there are multiple tariffs and billing systems between and within different ERIL operators. Unlike in the concessionary areas, ERIL project operators have no recourse to compensation for charging affordable tariffs and many customers have been unable, and/or reluctant to pay (MPE and GIZ 2020:12).

Table 3. ERIL mini grid programmes. Adapted from Power Africa (2019). Credit: Off-grid Solar Market Assessment—Senegal.

Year built	Programme and donor	No. of minigrids	Status	Issues
1996	Project Nippon& Govt of Japan	3	No longer operational	Tariffs did not permit sustainable operation and maintenance (O&M)
2000	Project Convention & Govt of Senegal	53	No longer operational	O&M contracted not finalised
2005-2006	Project Isofoton & Govt of Spain	9	No longer operational	Undersized; unsustainable O&M
2008	PERACOD & Govt of Netherlands	18	No longer operational Undersized.	No metering
2010	PERACOD & Govt of Netherlands	70	Significant number no longer operational	Undersized; no metering
2011	Project Daey Ouwens & Government of Netherlands	10	Operational	No major issues
2011	Autrichian Project/GoS	27	No longer operational	No O&M contract
2018	ECREEE & European Union	40	Not yet operational	Have been built, but O&M tender delayed
2020	European Union	70	Not yet built	Engineering, procurement and construction (EPC) tender delayed
2020	Islamic Development Bank, West African Development Bank, ECREEE, ASER	188	Not yet built	EPC tender delayed
2020	Project European Renewable Energy Federation, ASER	2	Under construction	EPC has been selected
2020	PUDC/UNDP	122	20 built, not yet operational	Remaining EPC not funded; O&M contract not yet funded
2020	Abu Dabi Fund & IRENA	30	Not yet built	Design study underway
2020	Gauff & Kreditanstalt für Wiederaufbau (KfW)	300	ASER 300. Under construction	Design study underway

Source: Power Africa (2019:11).

Despite attempts to create boundaries between ERIL concession areas and the three-year plans of the PPER concessionaires, there have been cases of overlap and competition between the two and provisions for grid arrival have not been properly established (USAID and Power Africa 2018:70). Some electricity consumers supplied by the ERILs live in neighbouring villages to those supplied by concessionaires, which has resulted in an inevitable comparison of tariff and service levels between these two groups and an awareness that those supplied by the ERILs are for the most part paying a higher tariff for a less reliable and less continuous supply of electricity than those supplied by the concessionaires. This has resulted in a reluctance to pay among many ERIL customers (MPE and GIZ 2020).

There has also been a mismatch between mini grid project capacity and project demand. ERIL's previous 200-household limit, which was eventually lifted with the introduction of the 2021 Electricity Code (Ba 2018:34), made it hard to generate sufficient customers for private sector actors to consider mini grids profitable, particularly given that many low-income consumers are dependent on seasonal agricultural revenues that do not correspond to monthly billing systems (Interview 3). The cap also posed a challenge in villages greater than 200 households and where electricity demand has sometimes

been higher than the project capacity of approximately 35 kw. In this case concessionaires have had to increase the capacity themselves and cover the cost of doing so. In addition, because of a lack of continuous reliable supply from many ERIL mini grid projects, which generate at approximately six hours per day and often less, much of the electricity generation has been directed to domestic needs as opposed to productive use such as millet mills, sewing machines, irrigation pumps and ice production. This curtailed supply has also led to a mistrust of the systems by users (MPE and GIZ 2020).

In light of ERIL's shortcomings, the 2021 Electricity Code created a new model with which to replace it called 'decentralised rural electrification' (L'électrification rurale décentralisée (ERD)), with the aim of creating economies of scale, attracting larger clients and being of greater interest to private developers. In addition to removing the 200-household limit, the Code also lifted the cap that limited mini grid project capacity to 1 MW (République du Sénégal 2021). The more recent installation of newer, third generation mini grids with battery storage which will have the capacity to operate at 24 h per day may lead to a more favourable perception among local communities. But there is as yet limited data on progress.

Additional actors within both the ERILs and the PPER also include SHSs which range from three light bulb systems to more complex set ups involving household goods such as TVs and fans and are distributed by companies such as Baobab +, Oulu Solar, BBOX and Sunergie. Rather than selling electricity in the form of kWhs, these companies distribute the kit and allow consumers to acquire and pay for the system over time. Under the current regulation, some SHS distributors are operating illegally within the concessions by competing with SHS tenders under the ERILs and the PPER. This is nonetheless tolerated by ASER and the concessions themselves (RVO 2022:23).

5.2.1. From ERSEN to ASER 300

Many ERIL projects were developed under a major programme called ERSEN⁷ I and 2, which ran between 2006 and 2017 and was subsidised by GIZ's EN-DEV programme, the Dutch PERACOD programme and the EU (Etienne 2022). Donors subsidised 70% of ERSEN's cost with the remainder coming from the operator and the relevant municipality (EnDev n.d.). The programme focussed on rural electrification of households, schools and health posts, as well as public lighting in 265 villages of up to 700 habitants through PV-diesel hybrid mini grids, or SHSs in smaller villages. Most of the villages were clustered in the regions of Sédhiou and Kaolack, and to a lesser extent in Fatick and Kolda (PERACOD n.d., see figure 3). In the case of mini grids, ERSEN covered the project's installation and infrastructure costs with operation and maintenance costs delegated to Senegalese small and medium operators. Significantly, ERSEN failed to adhere to the spirit of the ERIL programme in that rather than being initiated by local community organisations, it was top-down and donor led (Etienne and Robert 2024).

While many of Senegal's existing mini grids were installed under the ERSEN programme, of the approximately 96 solar diesel hybrid mini grids projects installed MPE and GIZ (2020:3), an estimated 52% are no longer functioning, meaning that they would have had an average lifespan of six years. Those that remain in operation generate electricity for an average of three hours per day, in contrast to the anticipated total generation of six hours per day (Semis 2020, in Etienne 2022). According to various interviewees, the reasons for this were poor operational performance, lack of management and maintenance, problems with commercial and financial management, and a lack of revenue generating activities from the projects themselves.

More specifically, interview 5 summarised, '[the ERSEN] programme installed the highest number of mini grids that currently exist in the country, but they also installed the highest number of mini grids that are currently not operating' and that 'the first phase of the ERSEN project was for mini grids of a relatively small capacity which had not necessarily taken full account of the needs of the electricity users concerned. Many projects were unable to satisfy the demand of the users which was considerably higher. The projects were being paid for on a fee for service basis, meaning that users would pay a certain amount each month depending on their electricity use'.

In the wake of ERSEN's shortcomings, a subsequent programme called ASER 300, was launched in 2020 under which ASER awarded a tender to German company Gauff Engineering for an engineering, procurement and project management contract for an additional 300 solar mini grids across the country (Get Invest 2021). The mini grids, all of which will be equipped with battery storage, have a generation capacity of between 15 and 45 kWp per village and will include the installation of distribution infrastructure and solar streetlamps (Gauff Engineering 2024; Takoulu 2023). A total of 24 000 domestic connections is anticipated, with five sockets and five bulbs included in each connected household. Some projects will also supply equipment for productive energy, including fridges, water pumps and grain

⁷ Acronym for: 'Electrification rurale Senegal'. The EN-DEV programme ran from 2005 and 2019 and as well as in Senegal, was also present in various countries in Africa, Asia and Latin America.

mills. The project also includes a training programme for local people for system installation and maintenance and is receiving finance from a subsidiary of German development bank KfW, with France's Euler Hermes providing financial credit insurance (Gauff Engineering 2024). After project completion ASER will transfer responsibility for operation, maintenance and the rights of revenue collection to a concessionaire or an ERIL, or if within Sénélec's jurisdiction, an independent operator, given the lack of interest from the utility in taking over mini grids.

To date, sixty of these solar mini grids, which are mainly for productive use, have been installed in Kolda, Upper Casamance, in southern Senegal, and 44 in the Kaffrine region on the Gambian border. Another 80 are being commissioned in the north of Senegal. In January 2020, ASER subsequently invited bids for the construction of mini grids in an additional 133 villages (Get Invest 2021). Importantly, these projects are larger than those installed under the first phase of ERSEN with the aim of demonstrating that with a minimum level of energy demand from productive users (up to 60%), the mini grid becomes much more economically sustainable.

5.2.2. Additional initiatives for rural electrification: PUDC

A further rural electrification initiative which has overlapped but not been coordinated with the ERIL programme is the Programme d'Urgence de Développement (PUDC) (USAID and Power Africa 2018:70), introduced in 2015 by the Presidency to continue the work of the PSE. Electrification is one of three components, the other two being roads and water. As interviewees explained, the PUDC has included the electrification of villages through low and medium voltage networks, the installation of distribution lines, as well as village-level mini grid projects marked for priority electrification outside of the ERILs. In the first three years the PUDC was financially supported and implemented by UNDP and has since received further technical and financial assistance from various bi-lateral and multi-lateral donors including India, Spain and Austria (Sonko 2017) and the African Development Bank (AfDB 2022).

However, findings suggest that the programme has lacked adequate regulation, has often been carried out without the involvement of ASER and 'has been installed without rules' (Interview 5). Other interviewees described an extreme lack of coordination and communication between the PUDC and existing rural electrification programmes, resulting in duplication in various villages and the need to remove recently installed mini grids. Another interviewee explained that the PUDC was politically motivated as part of election promises and failed to consider the needs and the nature of the electricity demand in areas where existing rural electrification efforts, including mini grids, were on-going. More extremely, another interviewee suggested that the programme was merely a corruption scheme.

SEforAll and AfDB (2019) added that the PUDC's design may have distorted the market given that mini grids built under the programme are built and handed over to ASER who is then in charge of transferring operation to a Transitory Delegated Operator (*Gestionnaire Délégué Transitoire*), or to the concessionaire of the area in which the project has been installed. While operators then hold the responsibility for ensuring the connection and operation of the systems, they do not have the ability to influence the ex-ante business models. Significantly, because ASER has been excluded from the running of the programme, it is unclear how many of the mini grids installed under the PUDC currently have connections in place, and whether or not ASER is monitoring them (SEforAll and AfDB 2019:11).

6. Discussion: key challenges summarised

Numerous reasons have been attributed as to why so many installed mini grids in Senegal are no longer operating, including a lack of advanced planning for long-term operation and maintenance; a failure to take account of patterns of local energy demand which has resulted in the under or over-sizing of the system; inaccurate data on the part of ASER on electricity access in some parts of the country; a lack of metering and monitoring of projects after their construction; and the lack of cost-reflective tariffs, which has ultimately deterred many investors (Signify Foundation 2019, Interviews 2 and 5). Further reasons that were common across all the rural electrification programmes discussed above are summarised below.

First, the arrival of the main grid into a locality where a mini grid is already established has often taken place in the absence of coordination and planning, not only between project operators and the distribution concessions, but also between operators and the PUDC. Though mini grid operators should be compensated if they are required to relocate their mini grid elsewhere, the provision of compensation has not always been forthcoming, suggesting a lack of clarity over the grid extension process, as well as an inherent conflict between the key institutions involved including Sénélec, some concessionaires, the PUDC, and mini grid operators (interview 7). Relatedly, the challenges of grid arrival for mini grid operators have been exacerbated by a problematic private licensing process given that while operators of

projects below 50 kW do not require a licence (SEforAll and BNEF 2020:84), they nonetheless require one to be able to claim compensation. Without this, should the grid arrive, operators have no claim on the site, and are unable to obtain loans to invest in major maintenance or upgrades (Power Africa 2019). Such circumstances have deterred potential investors and posed a challenge to the various programmes discussed above.

Second and as described by one interviewee, there is a vicious circle caused by the lack of regular technical oversight of the operation and maintenance of many mini grid projects, caused in part by a lack of a clarity in regulation and contracting between ASER and the technical operators. It is understood that operation and maintenance contracts are often short term, for instance two years, which means that the operator is often unable or unwilling to make financial investments in the system or even pay to mend any defects (interview 5). Without this long-term oversight, various component parts, particularly batteries, may start to malfunction or need to be replaced from about the third year of operation. But without the regular presence of technicians to prevent or resolve any issues, an absence of repair and maintenance results in system failure and the loss of electricity supply to the local community, who ultimately lose faith in the project and stop paying. A further issue, as previously discussed by Cholez and Trompette (2019:35), is that in the absence of clear systems for payment and for controlling use between the different households, some users have ended up cross-subsidising others.

Third, tariff setting for mini grids has long been problematic. For instance, as interviewee 2 explained in relation to the PPER, the tariff is often fixed on a monthly basis regardless of usage and regardless of whether the electricity supply has been consistent or reliable throughout that month. Moreover, there are various examples, particularly under the ERIL, of where mini grids have still been able to charge more than the harmonised tariff. So far, these tariffs have not been sufficient to cover the cost of operating and maintaining a mini grid and the cost of major technical upgrades, for instance to increase generation capacity, extend the distribution, or install meters.

Fourth, there is a lack of coordination and oversight, as much between donors as between state agencies involved in or responsible for electrification. As Interviewee 12 described: 'Some entities may be funded by the World Bank, others by the EU, or maybe there is a fund within the presidency that has been put there for political reasons. At the end of the day there is no true coordination in the sector'. Finally, there has been significant resistance from Sénélec to rural electrification initiatives which it perceives as a threat to its monopoly. As summarised by the World Bank (2013, 2013:8) 'Sénélec did not react well to the end of its monopoly in rural areas, making it reluctant to engage in the rural electrification project'.

7. Conclusion

As this paper has discussed, regulatory attempts to attract private investment into rural electrification in Senegal have been limited and made slow progress so far with most programmes being donor-funded. The various national institutions involved in rural electrification have been more focussed on grid extension than on decentralised options. The distribution concessions under the PPER meanwhile have for the most part been reluctant to embrace mini grids as an option due to the technical and economic risks. Despite the stated objectives to attract private investment, most mini grid projects installed under the ERIL and PUDC programmes have been donor-funded, with private participation largely limited to engineering, procurement and construction and operation and maintenance and ASER responsible for ownership. Mini grids have therefore made a relatively minor contribution to electricity access to date. With the exception of the ASER 300 programme which is relatively new, many of the projects that have been installed so far have been beset with technical, economic and/or social problems and are either no longer operational or suffering from poor performance. Such dynamics can be seen as an illustration of how global templates of rural electrification and poorly coordinated donor support have failed to align or take account of complex national realities and interests. With that in mind I offer the following four conclusions.

First, there are clear tensions in the electricity sector between state and private sector ownership; centralised and decentralised systems; and priorities for energy access and productive use. On the one hand, national regulatory frameworks and policies for mini grids and the participation of private actors are being put in place by government institutions responsible for rural electrification, particularly ASER and the regulator. On the other hand, there is still a strong national political discourse that favours the extension of centralised electricity and support to the utility Sénélec. This, in addition to an ambiguity within most of the concessionaires and/or a resistance by Sénélec towards off-grid electrification. Such tensions have been enhanced by multiple overlapping processes and institutions between whom coordination is not entirely clear, at once for the liberalisation of the power sector and for the regulation of

rural electrification and mini grids. This confusion has contributed to the regulatory uncertainty which appears to have discouraged investors under both the PPER and the ERIL programmes.

Second, because of the shortcomings of market-led rural electrification, most initiatives in Senegal, have been financed by public money, with grants, concessional loans and technical assistance from climate and development finance institutions playing a major role. Yet there are concerns regarding the longevity of such finance, and as flagged by SEforAll and AfDB (2019:11), ‘concerns persist on the ability of the government to continuously fund these projects, or to raise funds from international donors’. Moreover, as the case of ERIL, the PPER and the PUDC illustrate it is not just the *quantity* of funding that is of concern but also its *quality*, given that it does not always appear to have been well-spent, well-coordinated or well-targeted. There seems to have been limited accountability for donors involved in rural electrification in Senegal, who have continued to commit and disburse funds even when the programme has shown limited success and lacked coordination between and within national government and investors (interview 17).

Third, the case of Senegal challenges various assumptions regarding idealised models of power sector reform, energy access and rural electrification, mainly that the scale up of off-grid systems by private sector actors will automatically make electricity more affordable for the poorest users in geographies where access is limited or non-existent. Rather, in Senegal the shift towards private participation in rural electrification faces many challenges, not least the absence of a tariff structure that is deemed acceptable at once to investors, developers, the government, and consumers, and the lack of regulatory clarity over the extension of the centralised grid to areas in which mini grids have recently been installed. Further political economy and critical governance studies are therefore needed in order to understand the messy and constantly evolving politics of electricity sector reform within rapid technological changes.

Finally, the success of the concession approach to attract private investors to rural electrification initiatives has been challenged by the dispersed and remote nature of many rural populations who are low electricity users and are therefore not considered profitable. This reflects the incompatibilities between the financial expectations of project developers on the one hand and the precarious livelihoods and seasonal rhythms of many users on the other, who will not be able to meet expectations of profitability without either well-targeted public subsidies and/or business models that engage realistically with the opportunities and limitations present in rural agricultural value chains (Fajardo *et al* 2025, Sesan *et al* 2024b). As Cholez and Trompette *et al* (2019, 2019:339) argue, ‘how complex it appears to be to attempt to insert into calculation formulae both a private investor (with a capitalistic perspective) and an insolvent customer (who should become a better user and payer with the help of electricity as a development booster)’. The extent to which the ASER 300 programme, having learned from the shortcomings of the ERIL and PPER, may offer a more hopeful way forward, including through its training programme for local users and by increasing the size of mini grids to incorporate productive users is therefore an area for further research.

Data availability statement

The data cannot be made publicly available upon publication because they contain commercially sensitive information. The data that support the findings of this study are available upon reasonable request from the author.

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